

Step 1: logine to Trailhead Sealsforce -> Hands-on Orgs and launch the Playground

Step 2: Open Setup -> App Manager -> New Lightning App

App Name -> Student Management App 1

Step 3: Go to Setup → Object Manager → Create → Custom Object

Label -> Student Record

Plural -> Student Records

Record Name -> Student Name

Add Fields

- Student ID
 - o Type -> text
 - o Label -> Student_ID_1
 - o Unique

Step 4: go to Object Manager -> Create -> Custom object

Label -> Subject Record

Record Name -> Subject Name

Add Fields

- Marks
 - o Type -> Number
 - o Label -> Marks_1
 - o Length -> 3

Step 5: Object Manager -> Subject Record -> Fields & Relationship -> New

Select Master-Detail relationship

Fill

- Related To → Student Record
- Field Label → Student

Step 6: Create Total Marks

Go to Object Manager -> Student Record

Fields & Relationship -> New -> Roll-Up Summary

Fill

Field	Value
Label	Total_Marks_1
Object	Subject Record
Type	SUM
Field	Marks_1

Step 7: Create tabs

Go to Setup -> Search Tabs

Under Custom Object tabs -> new

Create

- Student Record tab
- Subject record tab

Step 8: Add to App

App Manager → Edit Student Management App 1

Add:

- Student Records
- Subject Records

Save

Step 9:

Object Manager → Student Record → Page Layouts

Open → Student Record Layout

From TOP panel (Fields):

Drag:

- Student_ID_1
- Total_Marks_1

Click Related Lists

• Drag:

+ ✓ Subject Records

Step 10: Launch app

Create Student

- Name -> Parth

- Student_ID_1 -> 101

Add Subject

- Maths 85
- Science 90
- English 80

And relate them to student Parth

Step 11: Add APEX CODE (Custom Function)

Go to Setup → Apex Classes → New

And paste write this code

```
public class StudentMarksController_Final {  
  
    public static Decimal calculateTotal(Id studentId) {  
  
        List<Subject_Record__c> subs = [  
            SELECT Marks_1__c  
            FROM Subject_Record__c  
            WHERE Student__c = :studentId  
        ];  
  
        Decimal total = 0;  
  
        for(Subject_Record__c s : subs) {  
            total += s.Marks_1__c;  
        }  
  
        return total;  
    }  
}
```

Step 12:

Validation Rule

Open Setup -> Object Manager -> Subject Record -> Validation Rules -> Click on New

Rule Name -> Marks_Limit_rule

Description -> Ensures marks do not exceed 100

Formula -> $\text{Marks_1_c} > 100$

Error Message -> Marks cannot exceed 100 Select

Field: Marks_1 and save

Step 13: Test Your validation

1. Go to your app
2. Open Student
3. Add Subject
4. Marks = 110 -> Error Shown
5. Marks = 95 -> Subject Stored